

Diagrammatic equation showing the addition of angular momentum states:

$$\begin{array}{c}
 \begin{array}{c} \square R \\ \square \\ \square \\ \square \\ \square \\ \square \end{array} \\
 \swarrow \quad \searrow \\
 jm \quad j'm'
 \end{array}
 +
 \begin{array}{c}
 \begin{array}{c} \square \\ \square \\ \square \\ \square \\ \square \\ \square R \end{array} \\
 \swarrow \quad \searrow \\
 jm \quad j'm'
 \end{array}
 =
 \begin{array}{c}
 \begin{array}{c} \square \\ \square \\ \square \\ \square \\ \square \\ \square \end{array} \\
 \swarrow \quad \searrow \\
 jm \quad j'm'
 \end{array}$$

The diagram illustrates the addition of two angular momentum states, jm and $j'm'$, to form a total angular momentum state R . The left side shows two separate paths (solid lines) leading to a common vertex, each associated with a state jm and $j'm'$. The right side shows a single path (dashed line) leading to a common vertex, also associated with a state R . The equality sign indicates that the two configurations are equivalent.